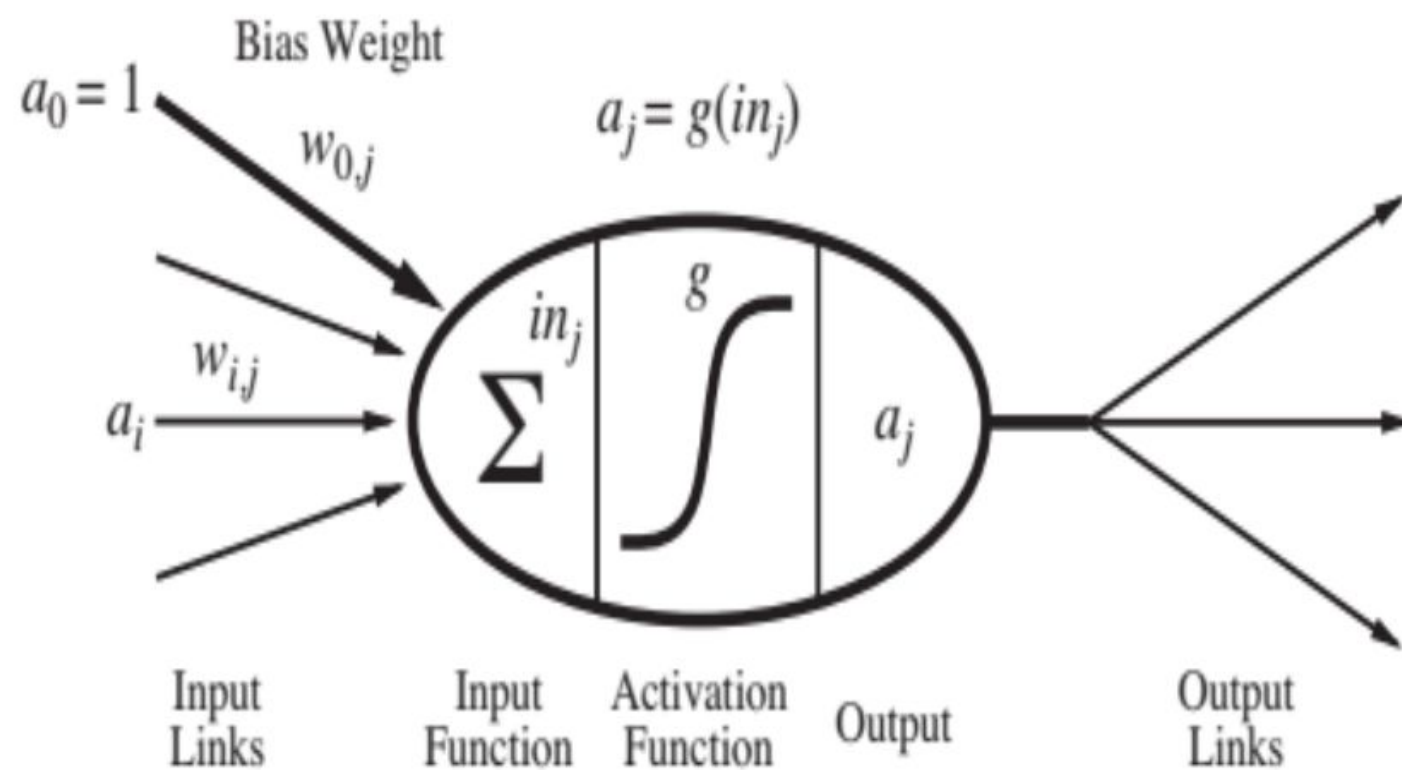


# Introduction to Neural Network



**Figure 18.19** A simple mathematical model for a neuron. The unit's output activation is  $a_j = g(\sum_{i=0}^n w_{i,j} a_i)$ , where  $a_i$  is the output activation of unit  $i$  and  $w_{i,j}$  is the weight on the link from unit  $i$  to this unit.

Neural networks are composed of nodes or **units** (see Figure 18.19) connected by directed **links**. A link from unit  $i$  to unit  $j$  serves to propagate the **activation**  $a_i$  from  $i$  to  $j$ .<sup>8</sup> Each link also has a numeric **weight**  $w_{i,j}$  associated with it, which determines the strength and sign of the connection. Just as in linear regression models, each unit has a dummy input  $a_0 = 1$  with an associated weight  $w_{0,j}$ . Each unit  $j$  first computes a weighted sum of its inputs:

$$in_j = \sum_{i=0}^n w_{i,j} a_i .$$

Then it applies an **activation function**  $g$  to this sum to derive the output:

$$a_j = g(in_j) = g \left( \sum_{i=0}^n w_{i,j} a_i \right) . \tag{18.9}$$

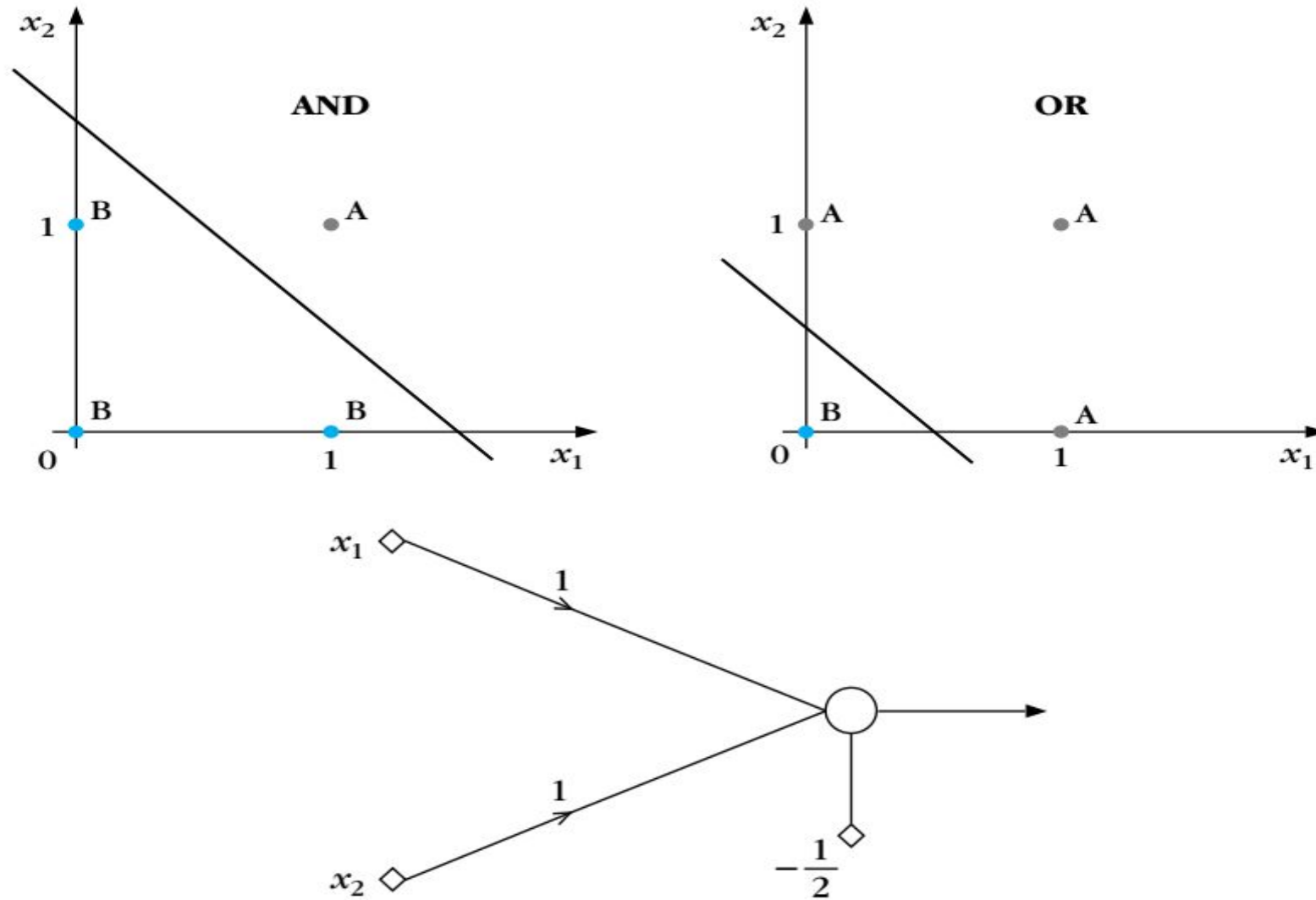
## Feedforward neural network

Directed acyclic graph. Only forward connection, no feedback loop. Output depends on current input and weights of the network

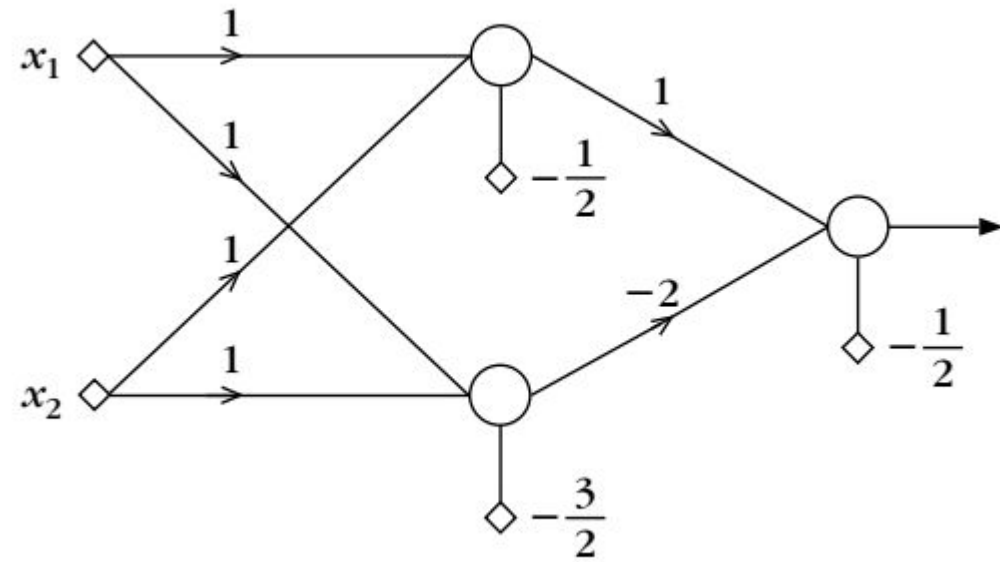
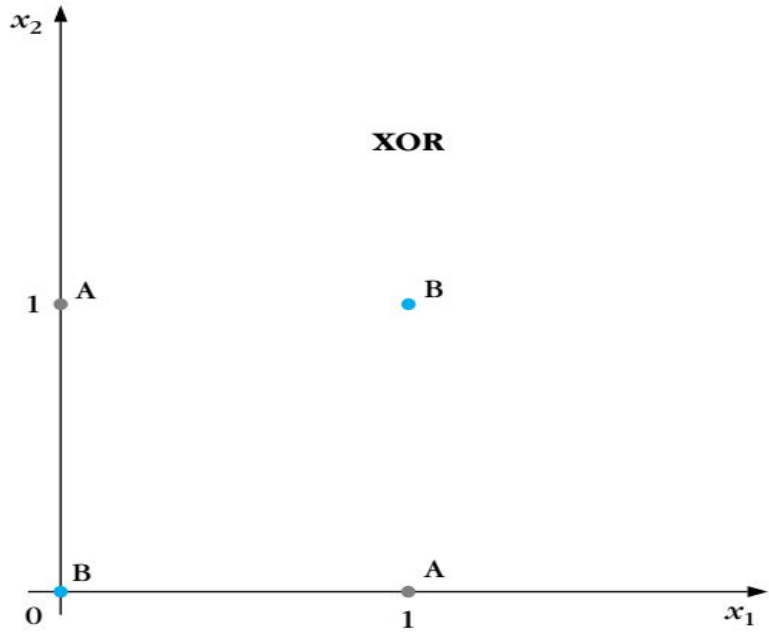
## Recurrent neural network

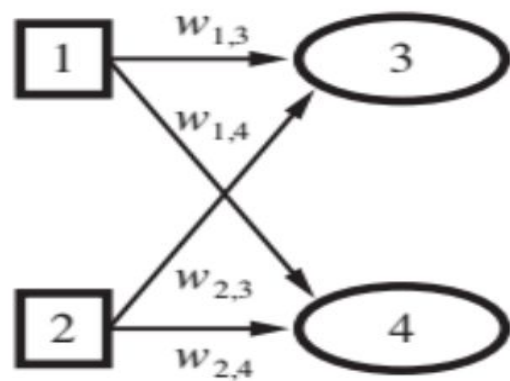
Feedback connection is available. Output depends on input and previous state. Suitable for short term memory

# Linearly separable function

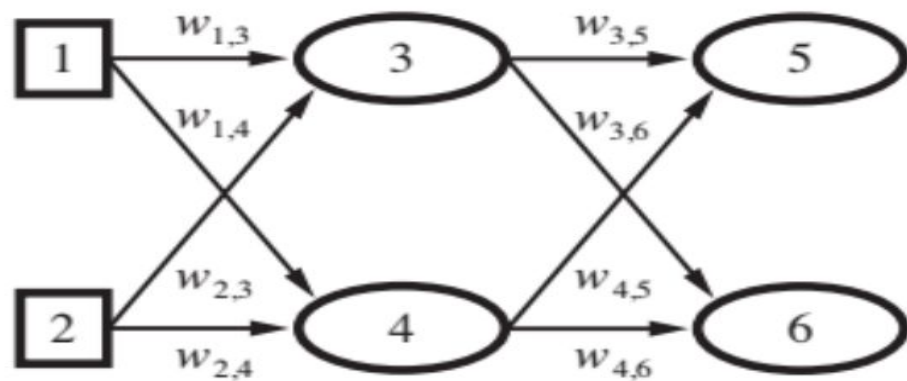


# Linearly non separable function





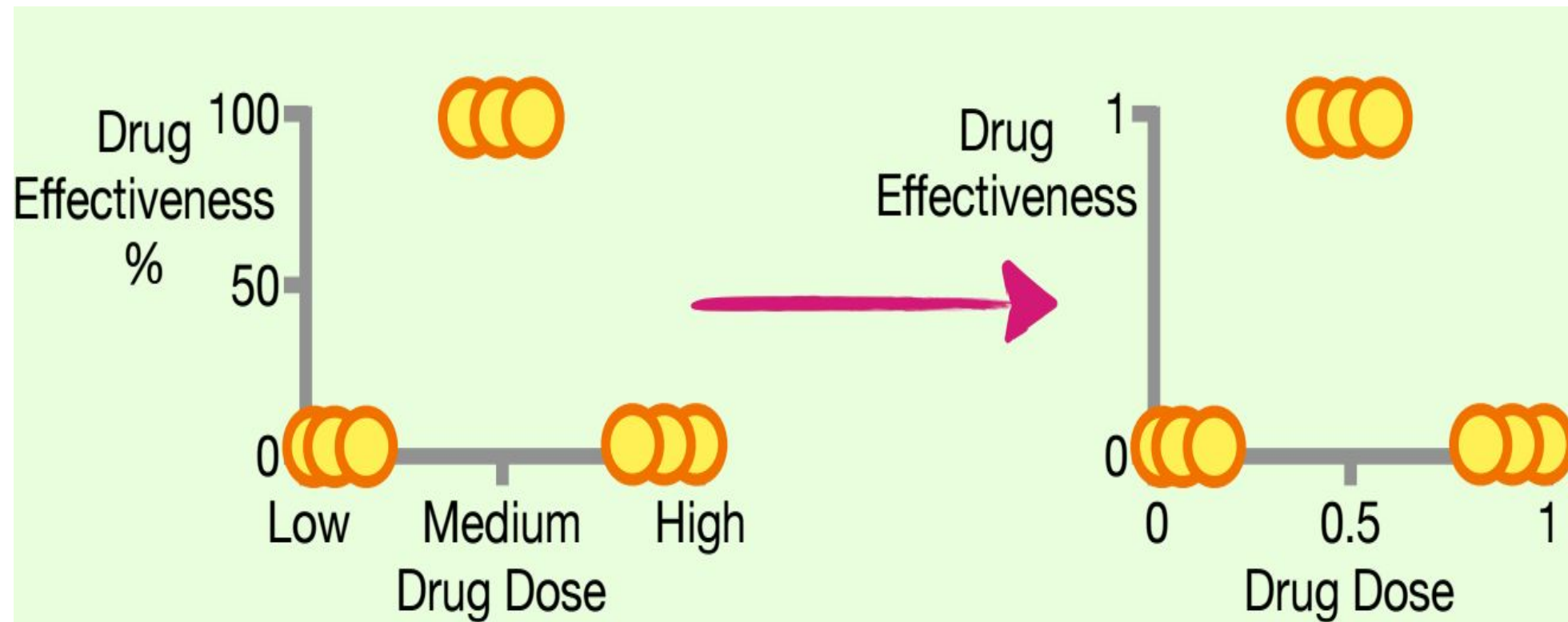
(a)

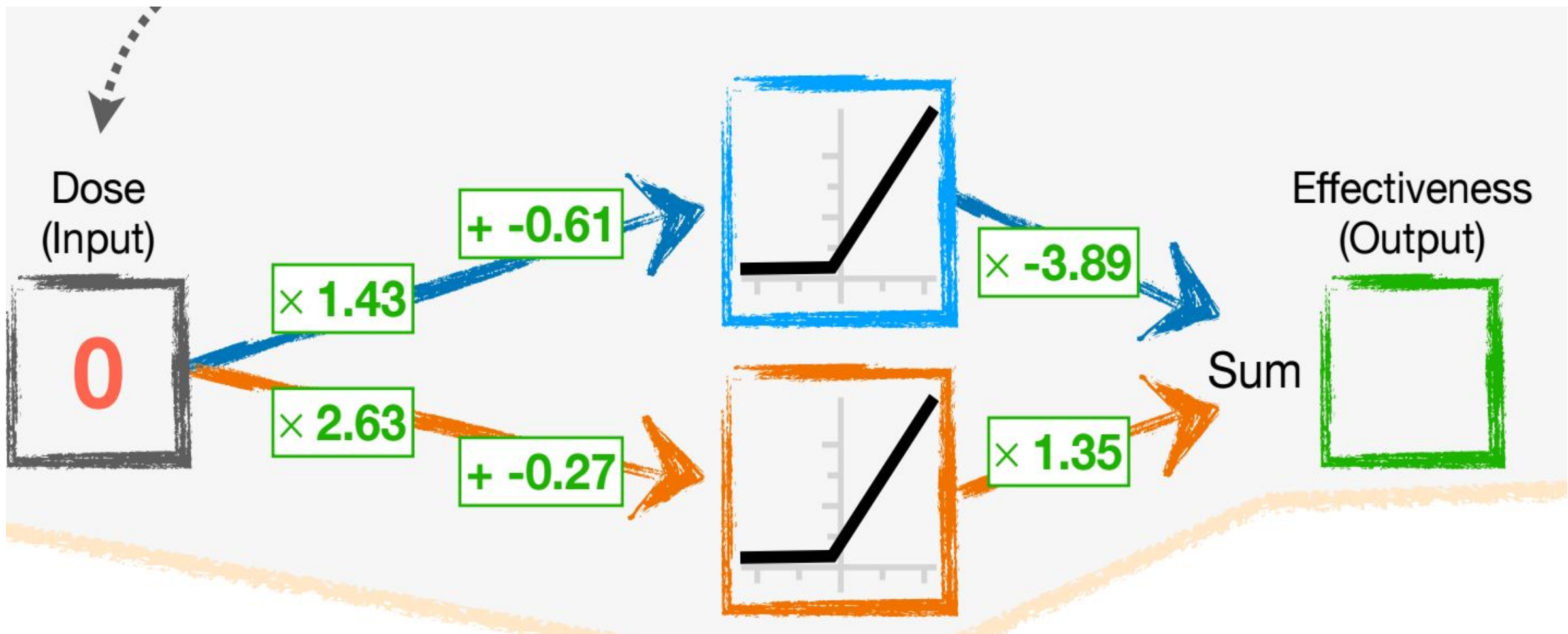


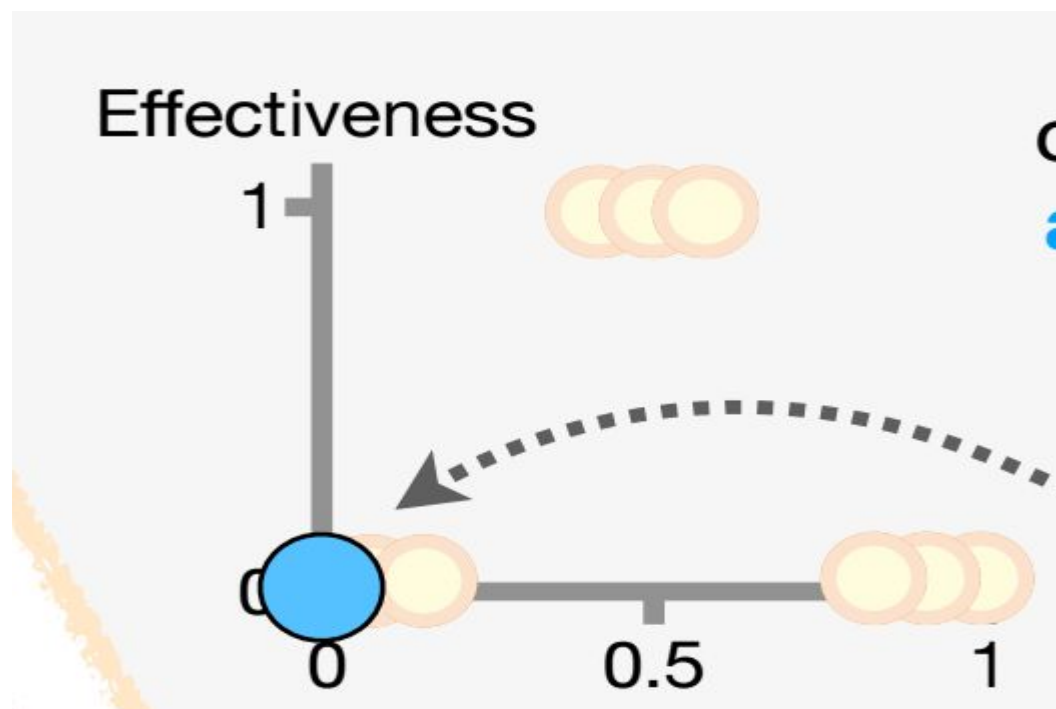
(b)

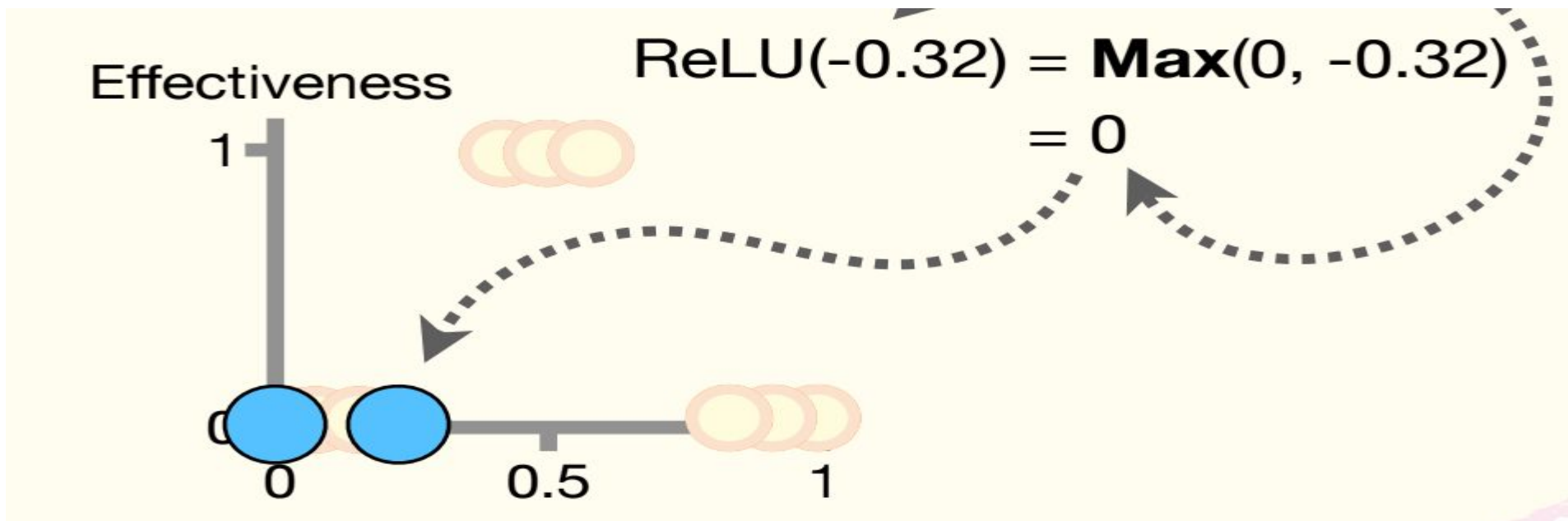
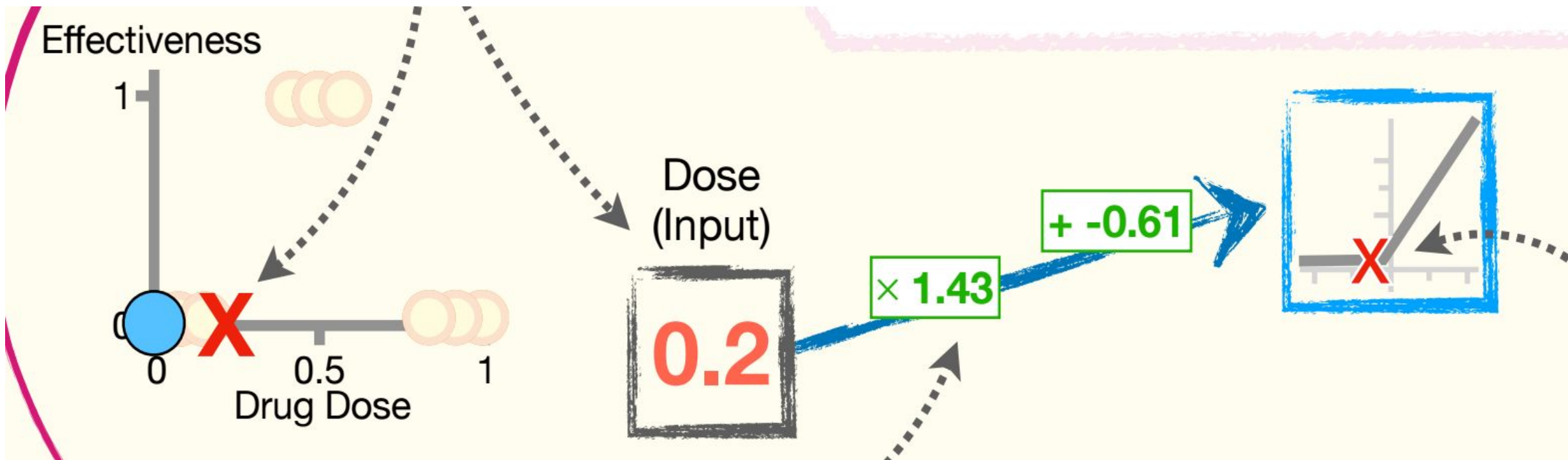
**Figure 18.20** (a) A perceptron network with two inputs and two output units. (b) A neural network with two inputs, one hidden layer of two units, and one output unit. Not shown are the dummy inputs and their associated weights.

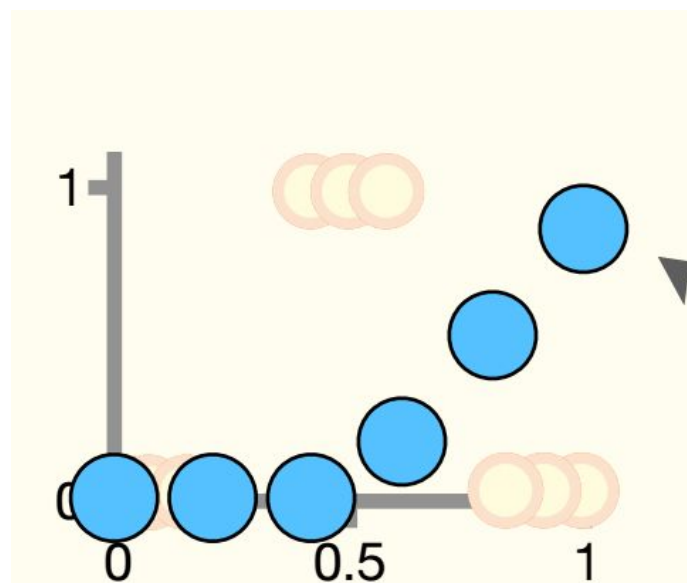
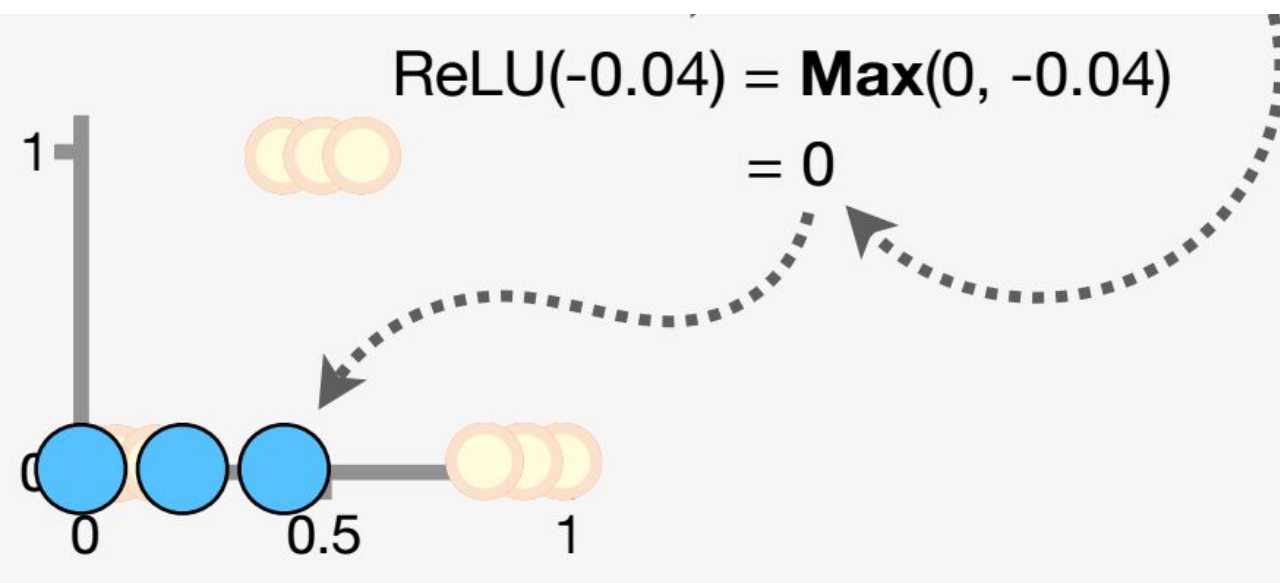
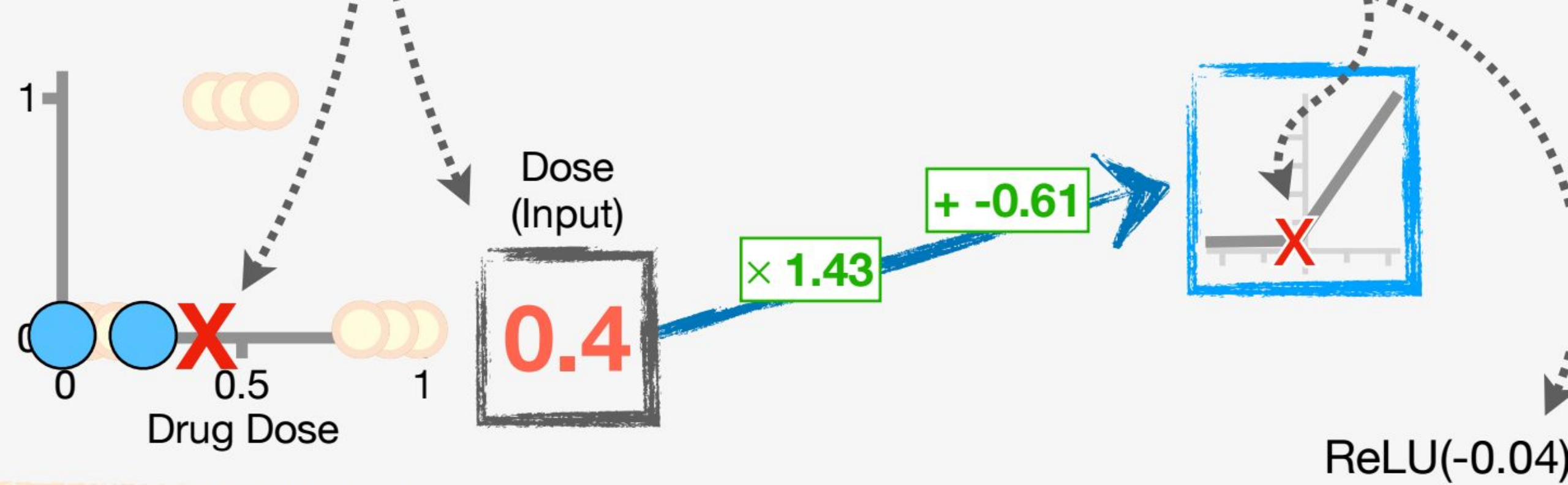
$$\begin{aligned}
 a_5 &= g(w_{0,5} + w_{3,5} a_3 + w_{4,5} a_4) \\
 &= g(w_{0,5} + w_{3,5} g(w_{0,3} + w_{1,3} a_1 + w_{2,3} a_2) + w_{4,5} g(w_{0,4} + w_{1,4} a_1 + w_{2,4} a_2)) \\
 &= g(w_{0,5} + w_{3,5} g(w_{0,3} + w_{1,3} x_1 + w_{2,3} x_2) + w_{4,5} g(w_{0,4} + w_{1,4} x_1 + w_{2,4} x_2)).
 \end{aligned}$$



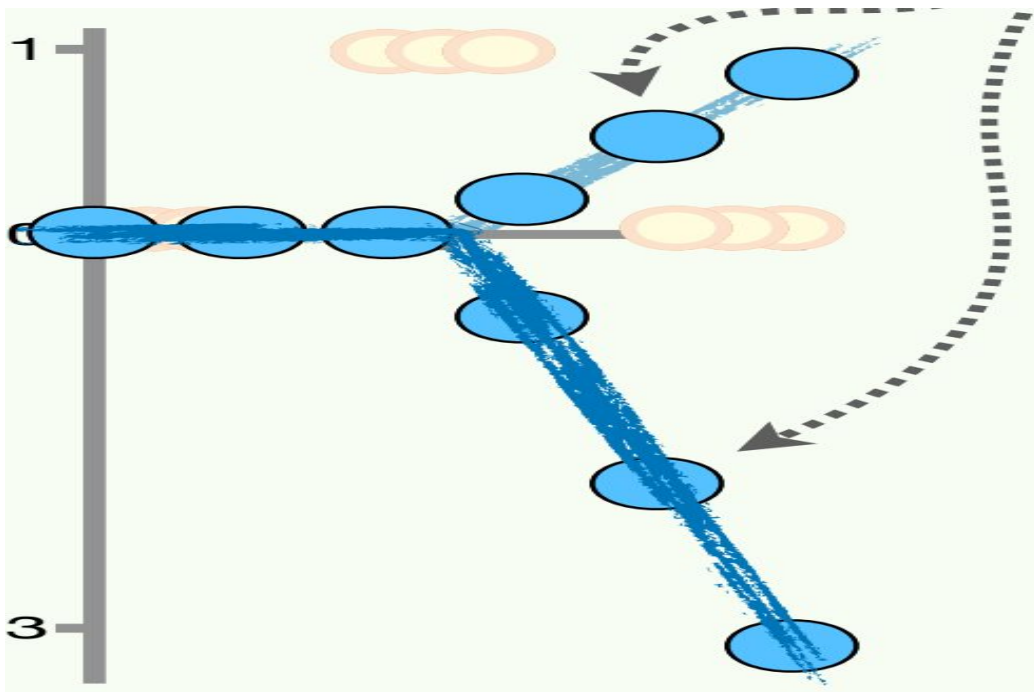
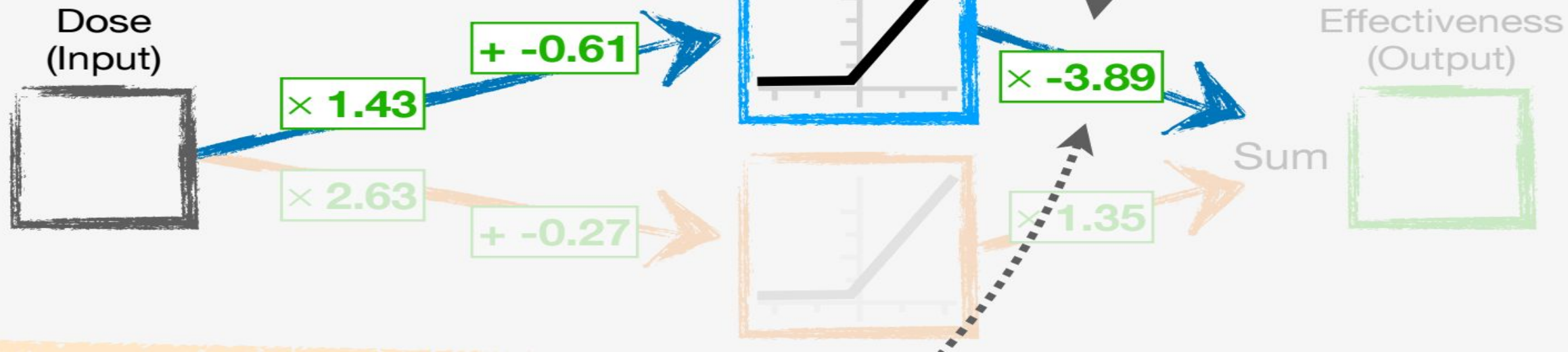


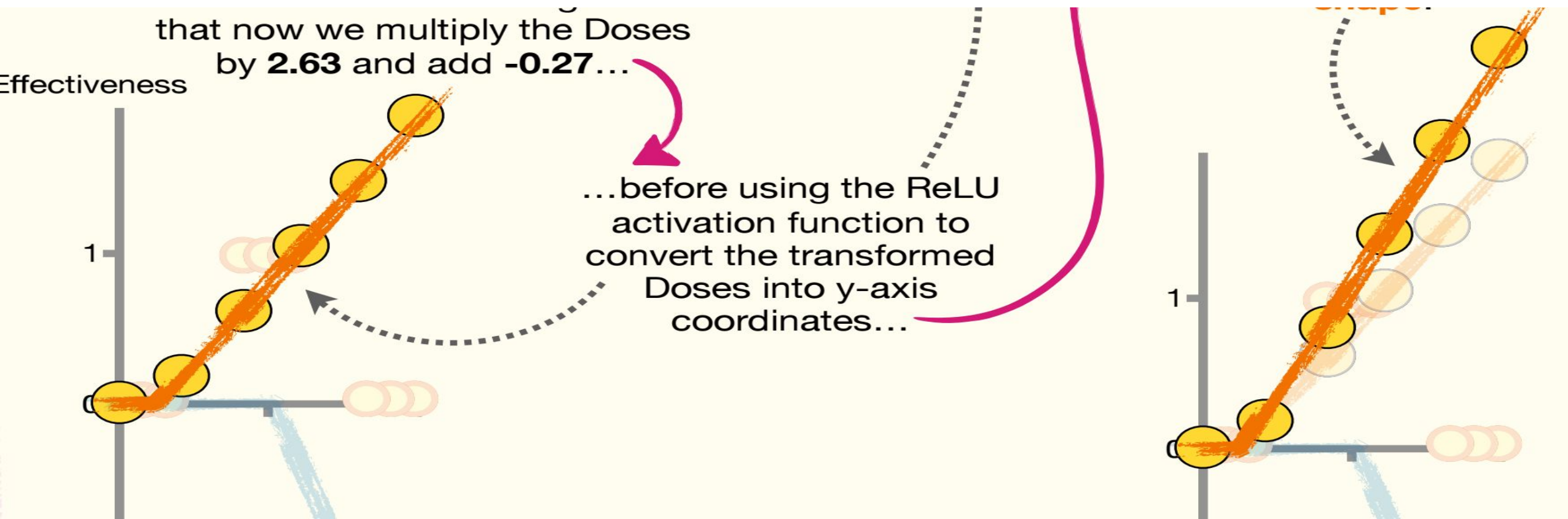


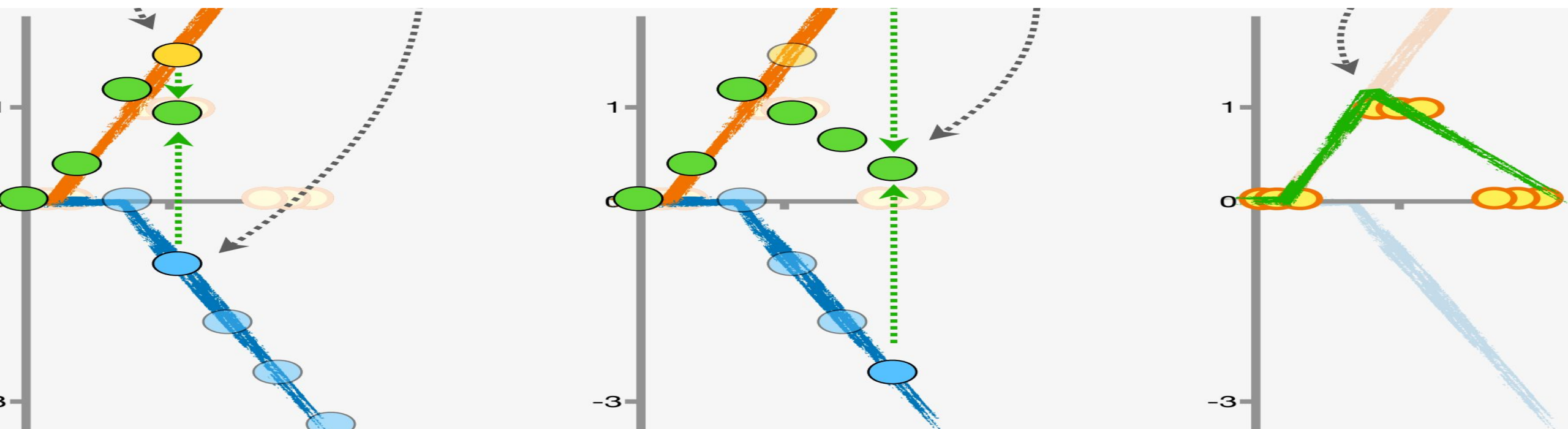
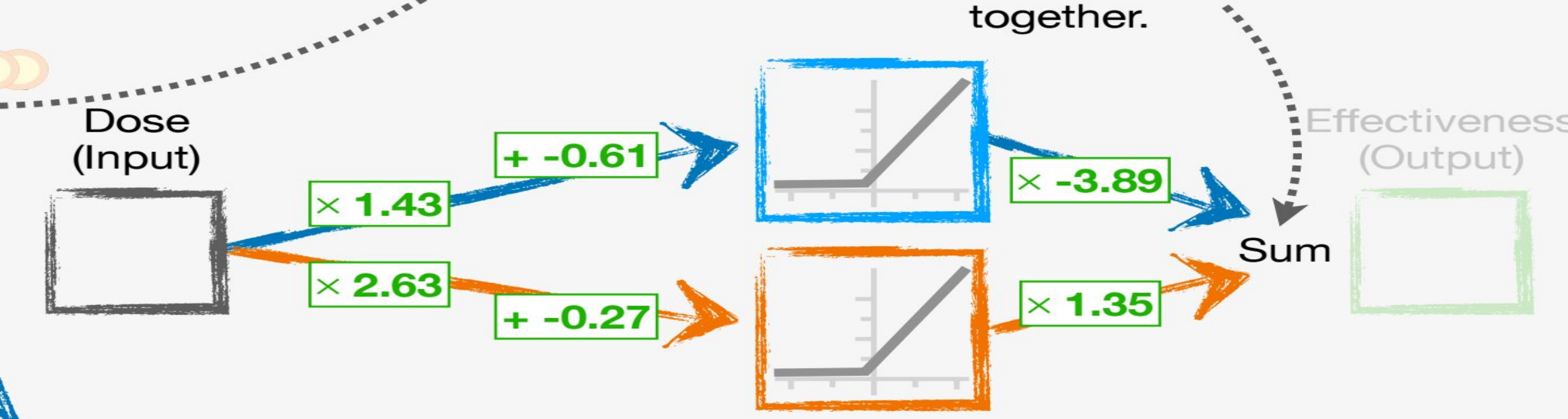


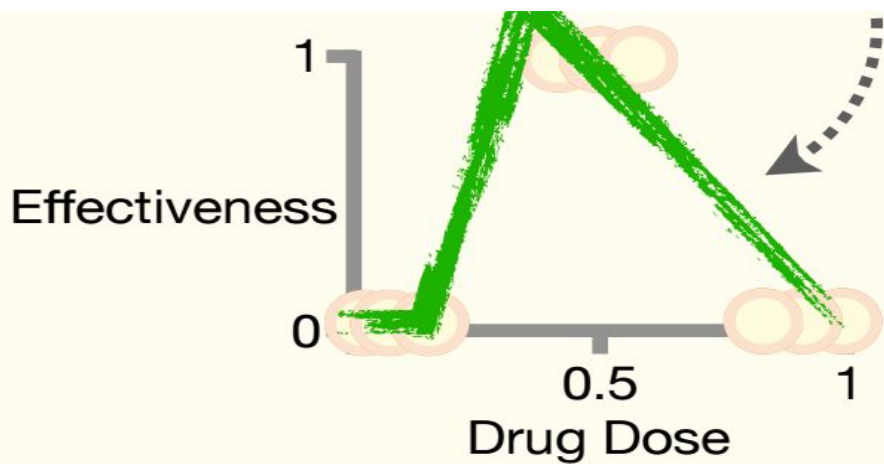


## Drug Dose









...that this neural network uses to predict Effectiveness from Doses.

